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Security Threats on the Electronic Bracelets for the Convicts in Jordan

**A project submitted
in partial fulfillment of the requirements for the
B.Sc. Degree in Cyber Security**

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CERTIFICATE

This is to certify that the project titled “ *Security Threats on the Electronic Bracelets for the Convicts in Jordan*”, submitted by the undersigned, is a record of original work carried out by them under my supervision, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Cyber Security**.

All the analysis, design, implementation, and system development work has been carried out by the undersigned. Furthermore, this project has not been submitted, either in whole or in part, to any other college or university for the award of any degree or diploma.

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ABSTRACT

Electronic bracelets are IoT-based devices widely used for monitoring and tracking individuals in applications such as healthcare, law enforcement, and the supervision of convicts. Because these devices depend almost entirely on the Global Positioning System (GPS) to report a wearer's location, they are exposed to serious cybersecurity threats — most critically GPS spoofing, in which falsified location data is fed to the device. Any successful spoofing attack can defeat the monitoring process, leading to privacy violations or system misuse, which makes verifying the authenticity of GPS data essential.

The objective of this project is to enhance the security and integrity of electronic monitoring systems by designing a machine-learning system capable of detecting GPS spoofing attacks in human-movement data. The project identifies the forensic signatures that distinguish genuine GPS readings from manipulated ones and uses them as the basis for automatic detection.

The methodology involves generating a dedicated, labeled human-movement GPS dataset that addresses the lack of publicly available electronic-monitoring data, engineering a rich set of movement-consistency features (such as the residual between reported and actual speed, satellite behavior, and course consistency), and training a soft-voting ensemble of Random Forest, Multi-Layer Perceptron, and Extra Trees classifiers. The system is evaluated using stratified, time-based, and blind testing on completely unseen data.

The results demonstrate that the proposed system distinguishes genuine from spoofed GPS signals with high reliability, achieving a blind-test accuracy of 99.58%. This project contributes a practical, accurate, and interpretable approach to securing electronic monitoring systems against GPS spoofing, and highlights the importance of data-driven cybersecurity in IoT systems with real-world impact.

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ABBREVIATIONS

- **IoT:** Internet of Things
- **GPS:** Global Positioning System
- **EM:** Electronic Monitoring
- **INS:** Inertial Navigation System
- **AV:** Autonomous Vehicle
- **CSV:** Comma-Separated Values
- **sklearn:** Scikit-learn
- **Agg:** Anti-Grain Geometry
- **ML:** Machine Learning
- **RF:** Random Forest
- **ET:** Extra Trees
- **MLP:** Multi-Layer Perceptron
- **NN:** Neural Network
- **HDOP:** Horizontal Dilution of Precision
- **GUI:** Graphical User Interface
- **PKL:** Pickle (serialized model file)

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CHAPTER 1: INTRODUCTION

1.1 Overview

Electronic monitoring bracelets are widely used in judicial and security systems to track individuals using GPS-based location monitoring. However, these devices are vulnerable to GPS spoofing attacks, where forged satellite signals mislead the bracelet into reporting a false location. This allows an individual to appear within an authorized area while actually being outside it, posing serious risks to public safety and reducing the reliability of electronic monitoring systems. This project focuses on analyzing how GPS spoofing attacks are performed, identifying the technical weaknesses of electronic bracelets, and improving spoofing detection mechanisms, and creating a realistic human-movement GPS dataset to enhance the accuracy and security of GPS-based tracking systems.

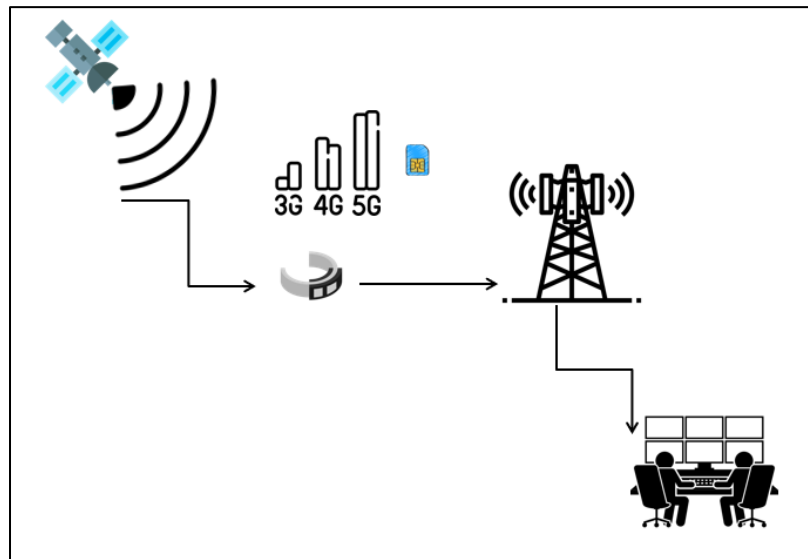


Figure 1 .Electronic Bracelet Architecture

The architecture illustrates the data flow and communication components of the electronic monitoring system:

1. GPS Satellite

GPS satellites transmit positioning signals used by the electronic bracelet to calculate the wearer's location.

2. Electronic Monitoring Bracelet

The bracelet receives GPS signals, determines the user's location, and prepares the data for transmission.

3. Cellular Network (3G / 4G / 5G with SIM Card)

Location data is transmitted from the bracelet to the backend system through cellular networks using an embedded SIM card.

4. Cellular Base Station

The base station relays the transmitted data from the bracelet to the central monitoring infrastructure.

5. Monitoring and Control Center

The control center receives and analyzes location data in real time, allowing authorities to verify compliance with movement restrictions.

Key Features for Spoofing Detection

1. **Speed Residual (speed_residual)**

The absolute difference between the reported GPS velocity and the speed calculated from coordinate changes. This is the most critical indicator of spoofing, as fake signals fail to maintain consistency between reported and actual movement.

2. **Satellite Locks (sat_locks)**

The number of satellites actively locked and used for positioning. Spoofed signals typically show abnormal lock patterns.

3. **Satellite Count (sat_count)**

The total number of visible satellites. Spoofing attacks often show inconsistencies with the number of locked satellites.

4. **Course-Bearing Difference (course_bearing_diff)**

The angular difference between the reported GPS course and the bearing calculated from consecutive positions. A significant discrepancy indicates potential signal manipulation.

5. **Coordinate Speed (coord_speed)**

Speed calculated from latitude and longitude changes over time. Serves as a ground-truth reference for validating reported velocity.

6. **HDOP (Horizontal Dilution of Precision)**

A measure of GPS accuracy. Spoofed signals often exhibit abnormal HDOP values.

7. **Course Change (course_change)**

The change in direction between consecutive readings. Spoofed trajectories often show unnatural smoothness or abrupt changes.

8.

Additional Features

The complete feature set (31 features) also includes: sat_ratio, sat_discrepancy, hdop_diff, velocity, velocity_diff, acceleration, distance_m, is_stationary, is_fast_human, along with rolling statistics (mean and standard deviation over a window of 5 readings) for velocity, coord_speed, speed_residual, sat_ratio, sat_discrepancy, hdop, course_change, and course_bearing_diff.

1.2 Project Motivation

The growing security challenges associated with Electronic Monitoring (EM) systems have created a strong need to ensure the reliability and integrity of GPS-based tracking technologies. Since EM bracelets rely heavily on GPS signals to monitor offender movement, they are vulnerable to GPS spoofing attacks that can falsify location data, affect

tracking accuracy, and reduce trust in monitoring programs. Current systems often lack advanced mechanisms for detecting abnormal GPS behavior or validating the authenticity of received coordinates, allowing spoofing attempts to go unnoticed. This project is motivated by the need to identify key vulnerabilities in EM bracelets and improve their resilience against GPS spoofing attacks. In addition, the lack of publicly available electronic bracelet GPS datasets encouraged us to collect real human-movement GPS data and use it to build a more practical detection approach. Ultimately, securing EM bracelets is essential for reliable offender supervision, public safety, and effective enforcement of legal restrictions.

1.3 Problem Statement

The integrity of Electronic Monitoring (EM) systems using GPS tracking bracelets is critically threatened by cybersecurity attacks that forge location data. The core vulnerabilities enabling these attacks include:

1. **GPS Signal Spoofing Attacks:** Adversaries can broadcast counterfeit GPS signals to deceive the tracking device, causing it to generate and report completely falsified location coordinates, altitude, and velocity.
2. **Data Replay Attacks:** Old, valid GPS data packets can be captured and re-transmitted at a later time to create a false record of the wearer's presence in a permitted location while they are actually elsewhere.
3. **Freeze Attacks:** The GPS position is frozen for an extended period with minimal noise added, making the device appear stationary while the wearer moves to a different location. This attack exploits the system's inability to detect unnatural position stagnation.
4. **Geofence Evasion Attacks:** The wearer is actually outside the permitted geographic area, but the system generates plausible walking patterns within the geofence

radius, with smooth transitions that avoid abrupt jumps. This attack manipulates both position and movement dynamics to appear legitimate.

These specific attack vectors exploit the system's blind trust in raw GPS data. Current solutions lack the ability to perform cross-feature forensic analysis on data packets to detect inconsistencies between multiple features such as reported velocity versus coordinate-calculated speed, course versus bearing, and satellite signal quality metrics. The detection system implemented in this project addresses these gaps by analyzing 31 features including speed residual, satellite locks, course-bearing differences, and HDOP anomalies to identify spoofing attempts.

1.4 Project Aim and Objectives

The main goal of this project is to deeply study, analyze, and find solutions for the security threats facing electronic monitoring bracelets used for convicts in Jordan, especially focusing on GPS spoofing vulnerabilities. To reach this main goal, the project aims to do the following:

Vulnerability Assessment and Threat Modeling: To find the security weak points in the electronic monitoring system, and analyze how attackers can intercept or mess with location tracking data.

Empirical Data Collection and Baseline Establishment: To connect theory with practice by building a hardware prototype. Using a Raspberry Pi and a NEO-6M GPS module connected via UART (at 4800 baud rate), we aim to collect raw NMEA sentences at 1Hz. This creates a solid dataset of normal, real human movement over 13 hours and 20 minutes.

Realistic Attack Simulation: To simulate smart and realistic GPS spoofing attacks—specifically Geofence Evasion, Replay, and Freeze attacks—to see how the system handles attacks. By splitting the movement data into separate sessions and rounds, our simulation uses four phases (Capture, Takeover, Stable, and Release) to make sure the movement changes smoothly and logically without sudden jumps in coordinates.

Adversarial Fine-Tuning for Realism: To make sure the fake attacks look like real ones, we dynamically adjust things like velocity, course, HDOP, and satellite lock metrics to mimic a weak or manipulated signal.

Machine Learning-Based Detection Framework: To develop, train, and test machine learning models that can accurately detect these fake manipulations and isolate them, making electronic bracelet tracking much more reliable and secure.

1.5 Project Limitations

- Several limitations were encountered during the development of this project:
- No publicly available electronic-bracelet dataset exists, and access to real data used by Jordanian authorities was not possible. Therefore, a dedicated human-movement GPS dataset was generated for this study.
- The spoofing attacks were synthetically generated and may not represent all real-world attack scenarios.
- The actual architecture and communication mechanisms of the electronic bracelet system used in Jordan are not publicly available.
- The proposed design is based on electronic-monitoring architectures used in other countries, which may differ from the system deployed in Jordan.
- The system operates on recorded datasets rather than live GPS streams and was not tested on embedded bracelet hardware.
- The project focuses on detecting GPS spoofing attacks rather than preventing them.
- Limited research resources and publicly available information on both GPS spoofing and electronic-monitoring systems constrained the study.

1.6 Project Schedule

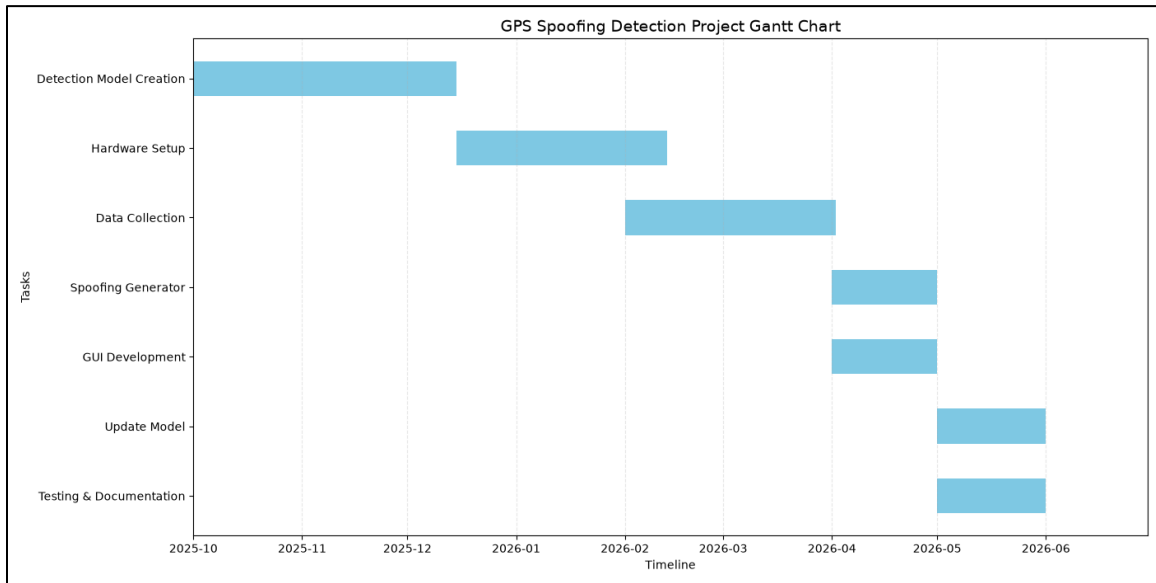


Figure 2 .Project Gantt Chart

1.7 Report Organization

Table 1 . Report Organization

Chapter	Outline
Chapter 2	Introducing literature review
Chapter 3	List the requirements analysis , including stakeholder identification and use case diagrams.
Chapter 4	outlines the system architecture and design, covering both software structure and UML diagrams.
Chapter 5	Implementation plan components

Chapter 6	Provides the testing plan
Chapter 7	The future work and concludes the report

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

This research builds upon a comprehensive investigation into the security threats targeting electronic monitoring bracelets used for convicts, with a specific focus on GPS spoofing vulnerabilities. The initial phase of our work involved analyzing potential system vulnerabilities and identifying the critical features required to distinguish between normal and malicious trajectories. This led to the development of an anomaly-based detection framework utilizing various machine learning models, which were initially evaluated using our detection framework to identify and isolate these adversarial manipulations. Using an autonomous vehicle (AV) dataset representing vehicular movement. To empirically validate and extend these findings to real-world scenarios involving human movement, we deployed a physical hardware prototype utilizing a Raspberry Pi and a NEO-6M GPS module operating via UART serial communication to capture raw NMEA sentences at 1Hz, as electronic bracelet trajectory data was unavailable. This chapter details our methodology for gathering a baseline dataset of genuine human movement over an extensive period of 13 hours and 20 minutes (comprising over 47,000 records across distinct sessions). Furthermore, it outlines our approach for simulating sophisticated and highly realistic GPS spoofing attacks—specifically targeting Geofence Evasion, Replay, and Freeze attacks. To ensure seamless and logical trajectory shifts without abrupt coordinate jumps, the simulation implements a four-phase transition model (Capture, Takeover, Stable, and Release). Finally, we elaborate on the dynamic fine-tuning of signal metrics to reflect degraded conditions, and the subsequent application of our detection framework to identify and isolate these adversarial manipulations.

2.2 Existing Systems

A variety of systems have been developed for monitoring and tracking the offenders. Some notable examples include:

Table 2 .Existing Systems

System	Description	Strengths	Weaknesses
Security of GPS/INS Based Location Tracking Systems [3] (2019)	System relies on the integration of GPS with INS to improve the accuracy of position tracking especially in areas where GPS signal is weak by detect the fake gps signals.	Detects GPS spoofing attacks. Improves location reliability. Does not rely on GPS alone	Not designed for electronic monitoring
Privacy-Preserving GPS-Based Electronic Monitoring System [4] (2024)	Monitors offender–victim proximity while protecting the victim’s location privacy.	Preserves victim privacy. Supports dynamic exclusion zones	no gps spoofing detection Complex system design
Safe Step Criminal Tracking System [2] (2013)	Tracks offenders in real time and analyzes their behavior and interactions	Real-time GPS tracking. Detects zone violations. Analyzes behavioral patterns. Raises alerts for possible reoffending.	Assumes GPS data is always correct. No GPS spoofing detection. Limited privacy considerations

2.3 Overall Problems of Existing Systems

- **Over-reliance on GPS data without validation:** Many existing systems trust raw GPS data without verifying its authenticity, making them vulnerable to GPS spoofing attacks and inaccurate location tracking.
- **Lack of bracelet-specific spoofing detection:** Most detection methods are designed for general GPS applications rather than electronic bracelets, limiting their suitability for wearable monitoring systems.
- **Scarcity of dedicated datasets:** Publicly available datasets specific to electronic-monitoring systems are extremely limited, forcing researchers to rely on substitute datasets that may not accurately represent human movement.
- **Limited feature analysis:** Existing approaches often use only basic GPS attributes and fail to analyze deeper movement-consistency patterns that can reveal spoofed data.
- **Unrealistic evaluation methods:** Many studies evaluate their models on data similar to the training set, without testing on completely unseen data, which may overestimate real-world performance.

2.4 Overall Solution Approach

This project focuses on improving the security of electronic monitoring bracelets against GPS spoofing attacks by developing a machine-learning-based detection system. The system analyzes human movement patterns and identifies abnormal behaviors that indicate manipulated GPS signals.

Due to the lack of public electronic-monitoring datasets, a labeled human-movement GPS dataset was generated containing both normal and spoofed records. The data is preprocessed by cleaning GPS records, handling missing values, and standardizing coordinates and time information.

Instead of relying only on raw GPS data, movement-consistency features are extracted, including speed differences, distance-based calculations using the Haversine formula, satellite information, course inconsistencies, acceleration, HDOP changes, and time-based movement patterns. These features help distinguish normal movement from spoofing attacks such as drift, freezing, replay, and signal manipulation.

The detection model uses an ensemble learning approach combining Random Forest, Multi-Layer Perceptron (MLP), and Extra Trees classifiers to improve accuracy and

reliability. The system is evaluated using stratified testing, time-based testing, and blind testing on unseen data to ensure realistic performance.

The proposed system provides a practical approach for detecting GPS spoofing in electronic monitoring systems, while real-time deployment on bracelet hardware remains a future improvement.

CHAPTER 3: REQUIREMENT ANALYSIS

3.1 Stakeholders

The stakeholders for the GPS Spoofing Detection System for Electronic Monitoring Bracelets are divided into Primary Stakeholders, who are directly involved in the operation and use of the system, and Secondary Stakeholders, who indirectly benefit from its functionality and insights.

Table 3 Primary Stakeholders

Stakeholder	Description
Judicial Police / EM Supervision Unit	The direct end-users who rely on packet classification (Legitimate/Spoofed) to make legal decisions and ensure the effectiveness of the monitoring program.
Tracking System Operators	Manage the technical system, supervise the detection model's operation, and monitor daily alerts and reports.
Cybersecurity Team in Judicial Institution	Analyze detected attack attempts, develop enhanced defense strategies, and integrate findings into broader security policies.

Table 4 .Secondary Stakeholders

Stakeholder	Description
Monitored Individuals (Offenders)	Indirectly benefit from a more fair and reliable system that ensures data accuracy and prevents false accusations from spoofed data.

Ministry of Justice / Legislative Bodies	Benefit from statistical reports on spoofing attempts to improve legislation and policies regarding alternative sentencing.
EM Bracelet Technology Providers	Can use insights about discovered vulnerabilities to enhance the security of their future device designs and firmware.

3.2 Use Case Diagram

The system use-case diagram is illustrated in Figure 3.

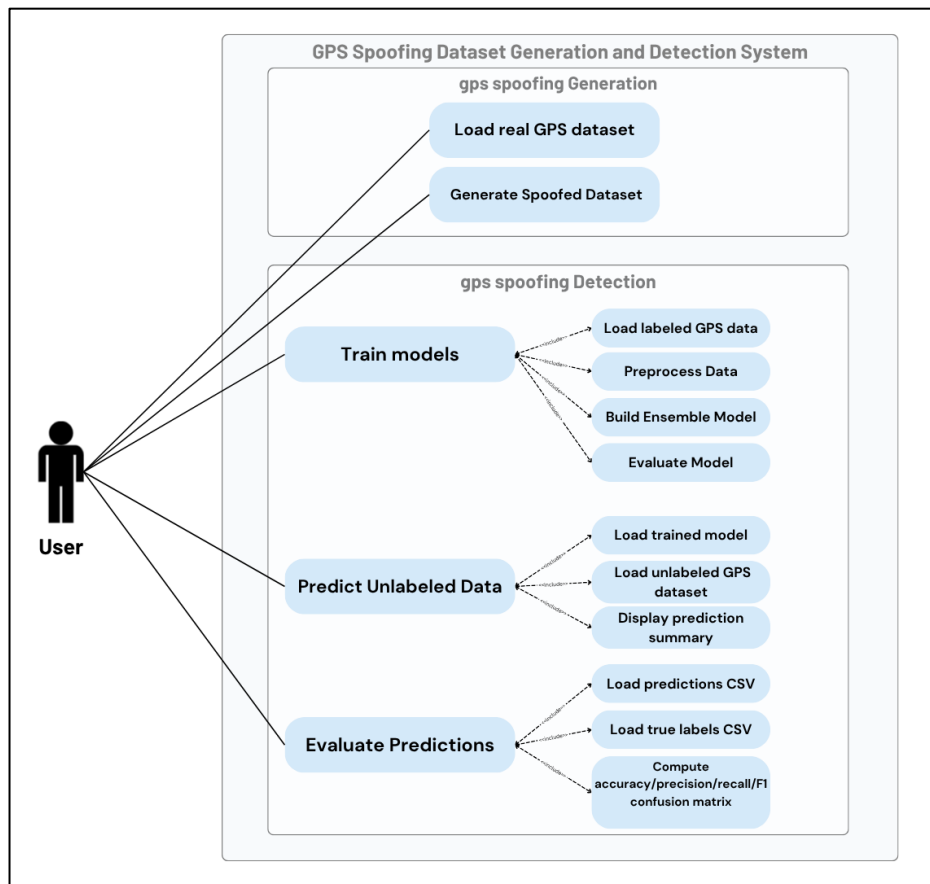


Figure 3 .Use Case Diagram

The use case diagram represents the main user interactions with the complete GPS spoofing detection system. It includes an offline dataset generation stage, where real GPS data is used to generate spoofed records, and a GUI-based detection stage, where the user trains models, predicts unlabeled GPS data, and evaluates prediction results.

3.3 Non-Functional Requirements

Accuracy & High Detection Precision: The system must achieve high precision in distinguishing between legitimate and spoofed data to minimize False Positives (falsely accusing a compliant individual) and False Negatives (missing a real attack). This is critical for the Judicial Police and Monitored Individuals to maintain legal credibility.

Performance & Real-Time Response: Classification and alert triggering must occur with minimal latency. This ensures that Tracking System Operators can intervene immediately when a violation is detected.

Transparency & Explainability: The AI models must provide interpretable results. This allows the Ministry of Justice and Legislative Bodies to understand the technical basis behind each alert for legal documentation.

Scalability: The architecture must be capable of handling an increasing number of monitored devices and larger datasets. This is essential for EM Technology Providers to integrate the solution into future large-scale infrastructures.

CHAPTER 4: ARCHITECTURE AND DESIGN

4.1 Software Architecture

The software architecture is illustrated in Figure 4.

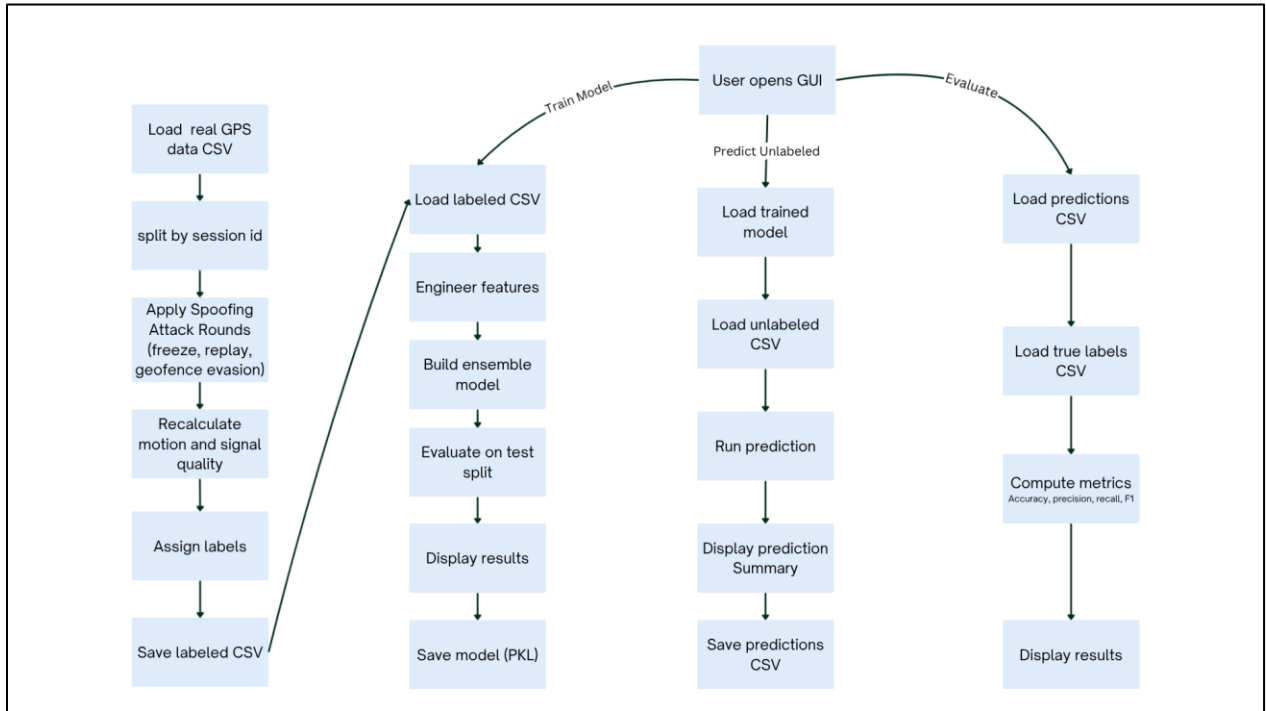


Figure 4 .Flowchart Diagram

The diagram shows the complete workflow of the proposed system. It starts by generating a labeled spoofed GPS dataset from real human-movement GPS data (collected using Raspberry Pi + Neo-6M) , then uses the GUI to train the detection model, predict unlabeled GPS records, evaluate predictions using true labels, and display the final results.

4.2 Software Design

4.2.1 Overview

The proposed system is designed to support GPS spoofing detection for electronic monitoring applications. It provides a complete workflow that starts with preparing a labeled GPS dataset and continues with model training, prediction, and evaluation through a graphical user interface. The system uses real human-movement GPS data, applies simulated spoofing scenarios for training purposes, and then uses machine learning models to distinguish between normal and spoofed GPS records.

The software design is divided into separate workflows because the main system actions are independent. Dataset generation, model training, prediction, and evaluation can each be performed separately depending on the user's needs.

4.2.2 Detailed Interaction

The main interactions of the proposed system are illustrated in Figures 5, 6, 7, and 8.

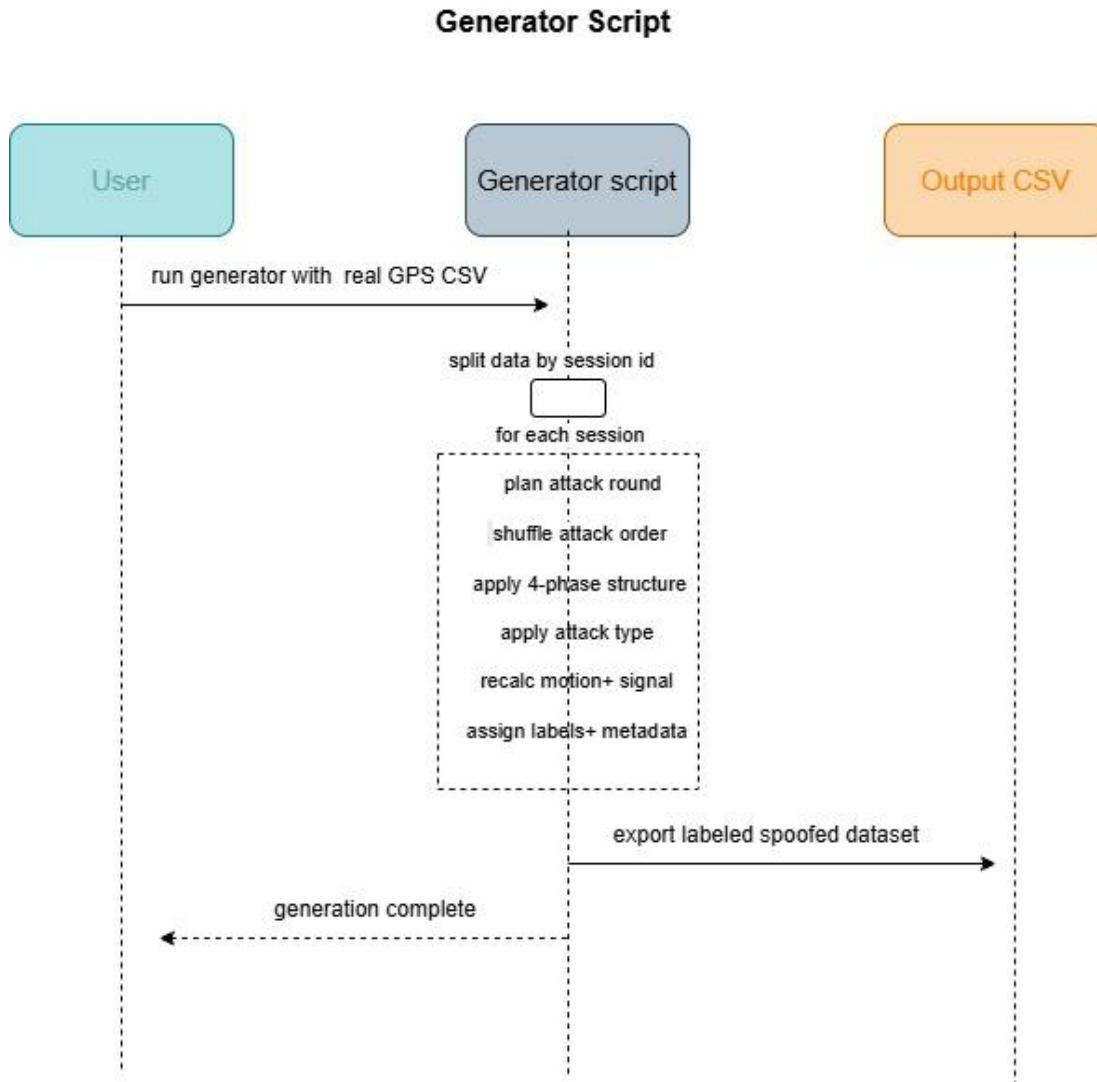


Figure 5 .Generator Script Sequence Diagram

This sequence diagram illustrates the workflow of the attack generator script. The process is initiated by the user running the generator script with the real GPS movement CSV file. The generator then splits the data based on session ID. For each session, it plans attack rounds, shuffles the attack order, applies a four-phase transition structure, and implements specific attack types while recalculating motion and signal metrics. Finally, it assigns malicious labels and metadata, exports the newly labeled spoofed dataset as an Output CSV file, and notifies the user upon.

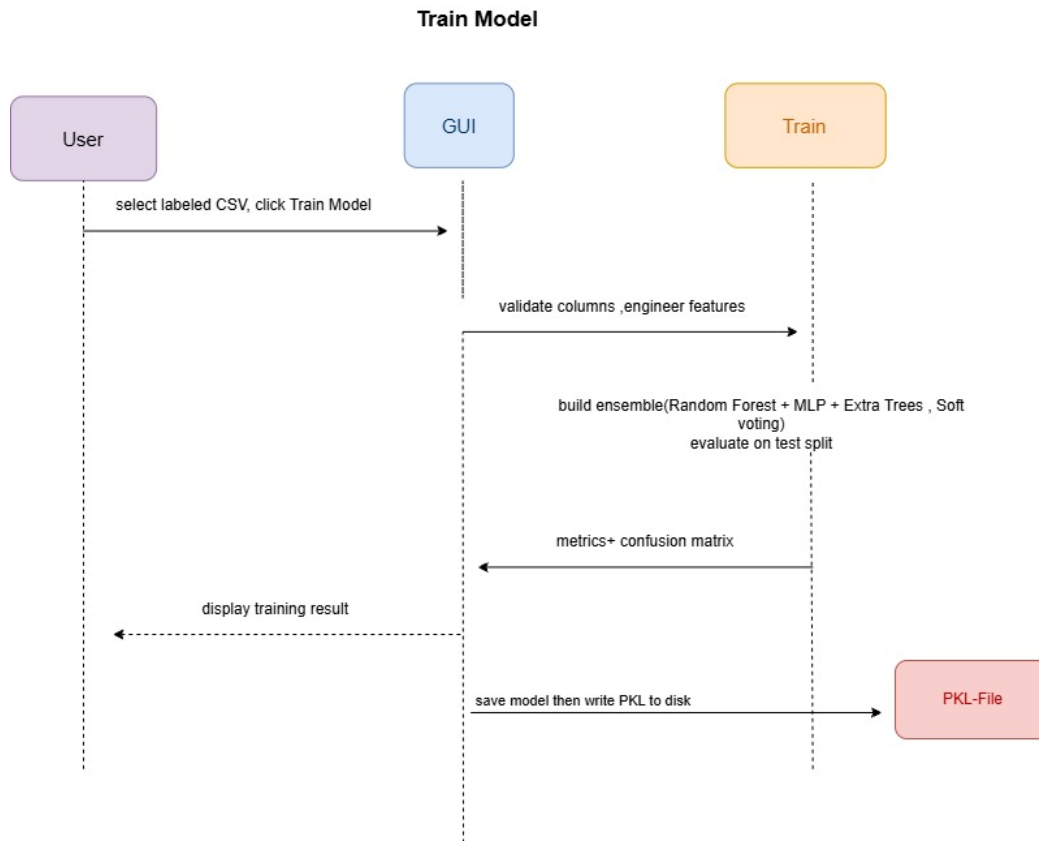


Figure 6 .Model Training Sequence Diagram

This sequence diagram illustrates the step-by-step process of training the anomaly detection models. The process begins when the user uploads a labeled CSV file and triggers the "Train Model" action via the Graphical User Interface (GUI). The GUI then validates the dataset columns and performs feature engineering. Next, the system builds an ensemble model combining Random Forest and Multi-Layer Perceptron (MLP) classifiers using a soft voting mechanism, and evaluates it on a test split. Finally, the training metrics and confusion matrix are displayed back to the user, and the trained

model is successfully saved and serialized into a .pkl file on the disk for subsequent real-time detection.

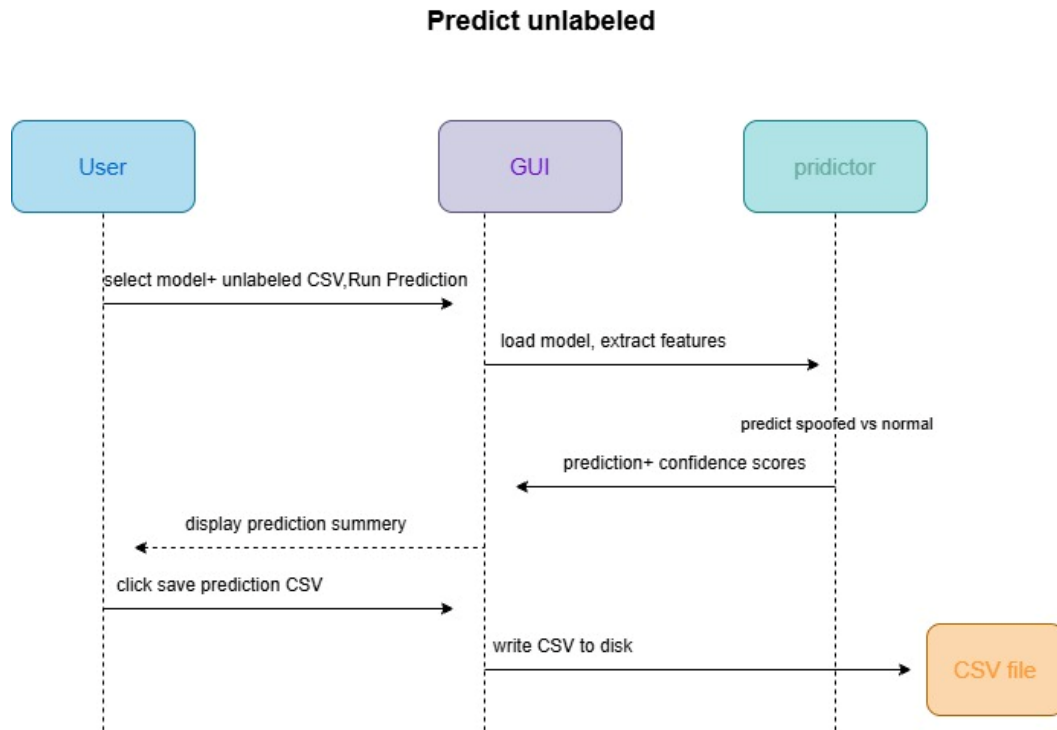


Figure 7 .Prediction Sequence Diagram

This sequence diagram illustrates the operational workflow for detecting GPS spoofing anomalies in real-time. The process begins when the user selects a previously saved model along with an unlabeled CSV file and triggers the "Run Prediction" action via the Graphical User Interface (GUI). The GUI loads the model and extracts the required features, sending them to the predictor component. The predictor then classifies the data into spoofed or normal, returning the prediction results along with confidence scores back to the GUI. A summary of the prediction is displayed to the user. Finally, the user can

choose to save the results, prompting the GUI to write and export the output as a new CSV file containing the predictions onto the disk.

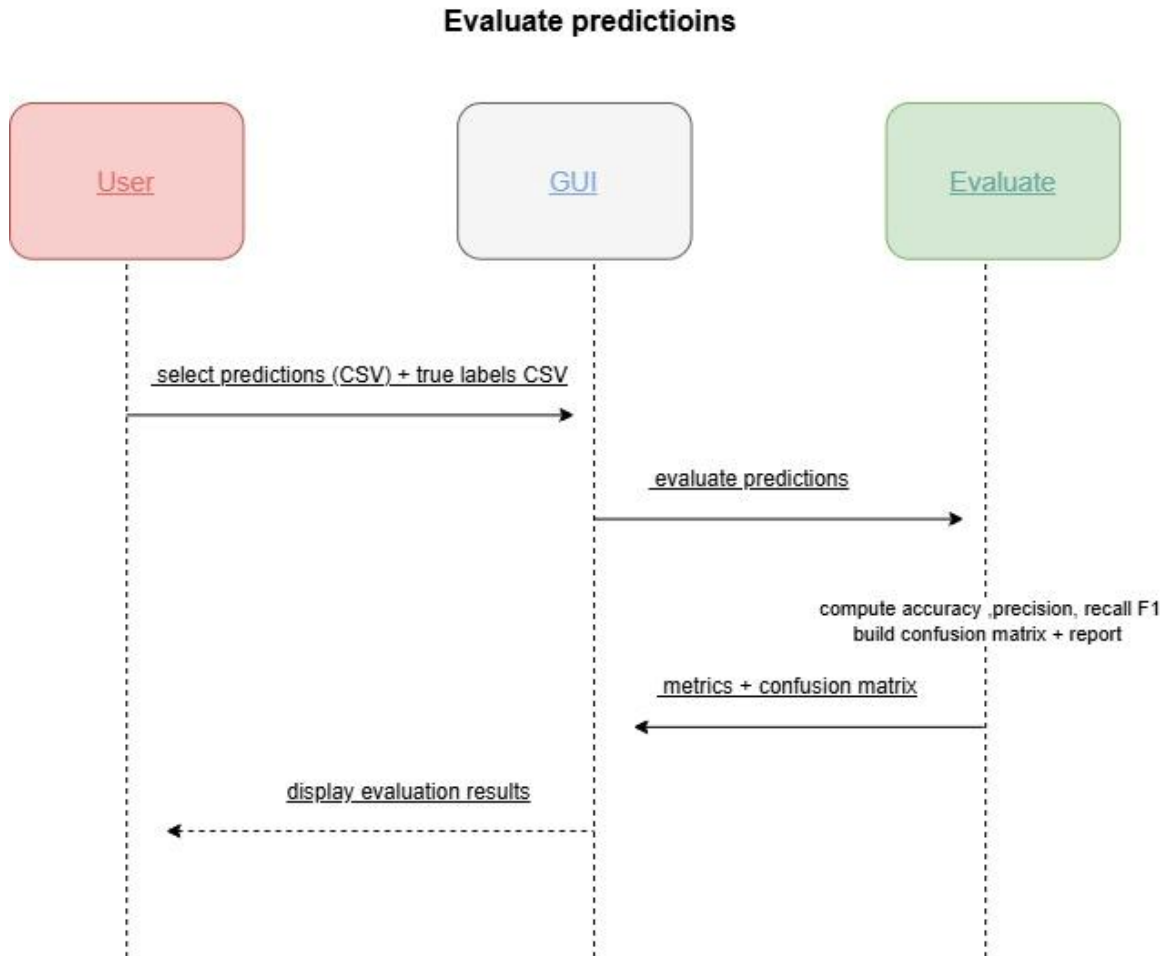


Figure 8 .Evaluation Sequence Diagram

This sequence diagram demonstrates the workflow for evaluating the model's performance on predicted data. The process starts when the user provides the generated predictions CSV file alongside the true labels CSV file through the Graphical User Interface (GUI). The GUI then forwards these files to the evaluation component to assess the predictions. The evaluator calculates critical performance metrics including accuracy, precision, recall, and F1-score and generates a confusion matrix along with a full classification report. Finally, these evaluation metrics are sent back to the GUI and displayed to the user for system performance validation.

Overall, Figures 5, 6, 7, and 8 show how the system supports dataset generation, model training, prediction, and evaluation as separate but connected functions within the GPS spoofing detection workflow.

CHAPTER 5: IMPLEMENTATION PLAN

5.1 Description of Implementation

During the implementation stage, a GPS spoofing detection system was developed using machine-learning models to classify GPS records as either normal or spoofed.

The main implementation steps included:

1. Dataset Preparation

Due to the lack of publicly available electronic-monitoring (EM) datasets, an Autonomous Vehicle (AV) GPS dataset was initially used as a substitute. Later, a dedicated labeled human-movement GPS dataset was generated to represent electronic bracelet data, containing both normal and spoofed GPS records in CSV format.

2. Software Setup

The required Python libraries were installed, including data-processing, visualization, machine-learning, and GUI development libraries. The system used scikit-learn components for preprocessing, model training, and evaluation, along with CustomTkinter for graphical interfaces.

3. Feature Engineering

Movement-consistency features were extracted from GPS data, including speed residuals, satellite information, course differences, acceleration, HDOP changes, and time-based movement patterns.

4. Model Development

Three machine-learning classifiers (Random Forest, Multi-Layer Perceptron, and Extra Trees) were trained and combined using a soft-voting ensemble approach. Data preprocessing was applied using imputation, scaling, and power transformation.

5. System Evaluation

The model was evaluated using stratified testing, time-based testing, and blind testing on completely unseen data to measure real-world performance.

6. Interface Development

Two graphical user interfaces were developed to allow users to operate the system without using the command line.

5.2 Programming Language and Technology

1- Programming Language: Python

Python was selected for this project due to its simplicity, readability, and strong support for data analysis and machine learning. It provides essential libraries such as Pandas and NumPy for data processing, Scikit-learn for developing machine-learning models, and Matplotlib and Seaborn for visualization.

Additionally, Joblib was used for saving and loading trained models, while CustomTkinter was used to develop the graphical user interfaces. These tools enabled efficient development of the GPS spoofing detection system with reduced complexity.

2- Modules And Libraries

- pandas - Loading and processing CSV data and converting it to tables.
- Used to read data files, process columns, and create new features.
- numpy - Performing mathematical and statistical operations.
- Used to calculate mean, standard deviation, mathematical operations, and handle array operations.
- matplotlib - Creating charts and graphs.
- Used to create visualizations including confusion matrices, feature importance plots, model performance comparison, and trajectory plots.
- seaborn - Enhancing chart appearance.

- Used to create styled heatmaps for confusion matrices with improved visual aesthetics.
- `sklearn.ensemble` - Ensemble learning models.
- Used to implement `RandomForestClassifier`, `ExtraTreesClassifier`, and `VotingClassifier` for robust spoofing detection.
- `sklearn.neural_network` - Neural network models.
- Used to implement `MLPClassifier` (Multi-Layer Perceptron) for deep learning-based detection.
- `sklearn.tree` - Decision tree models.
- Used to implement `DecisionTreeClassifier` for interpretable baseline detection.
- `sklearn.linear_model` - Linear models.
- Used to implement `LogisticRegression` as a baseline classifier.
- `sklearn.model_selection` - Data splitting and validation.
- Used for `train_test_split` (75/25 split) and `StratifiedKFold` cross-validation.
- `sklearn.preprocessing` - Data preprocessing.
- Used to implement `StandardScaler` for feature normalization and `PowerTransformer` (Yeo-Johnson) for data transformation.
- `sklearn.impute` - Handling missing values.
- Used to implement `SimpleImputer` for filling missing values with median.
- `sklearn.metrics` - Evaluating model performance.
- Used to calculate `accuracy_score`, `confusion_matrix`, `classification_report`, and `precision_recall_fscore_support`.
- `joblib` - Model serialization.
- Used to save trained models and preprocessing objects as `.pkl` files for later use.
- `matplotlib.use('Agg')` - Enabling graph saving without display.
- Used to allow saving graphs in server environments without a GUI display.
- `warnings` - Hiding warning messages.
- Used to suppress unnecessary warnings and keep the output clean.
- `os` - Handling files and folders.

- Used to create output directories ('human_detection_outputs', 'plots', 'models') and check for file existence.
- datetime - Time and date handling.
- Used to timestamp output files and reports with the current date and time.
- threading - Multi-threading support.
- Used to run GUI operations in separate threads to prevent interface freezing.

3-Models

- **Random Forest Classifier**
 - **Role:** Primary tree-based model and source of feature-importance analysis.
 - **Configuration:**
 - n_estimators=100
 - max_depth=10
 - min_samples_split=5
 - min_samples_leaf=2
 - max_features="sqrt"
 - class_weight="balanced"
 - bootstrap=True
 - **Function:**
 - Builds 100 decision trees with controlled depth to avoid overfitting.
 - Generates the feature-importance analysis that identifies which features matter most.
 - Produces the feature_importance.png chart.
- **Multi-Layer Perceptron (MLP) — Neural Network**
 - **Role:** Captures non-linear relationships among features.
 - **Configuration:**
 - hidden_layer_sizes=(50,)
 - activation="relu"
 - solver="adam"

- max_iter=80
- early_stopping=True
- **Function:**
 - Uses a single hidden layer of 50 neurons with the ReLU activation function.
 - Stops training early to prevent overfitting.
 - Serves as one vote in the final ensemble.
- **Extra Trees Classifier**
 - **Role:** Adds diversity through extreme randomization.
 - **Configuration:**
 - n_estimators=150
 - max_depth=12,
 - min_samples_split=5
 - min_samples_leaf=2
 - max_features="sqrt"
 - class_weight="balanced"
 - **Function:**
 - Builds 150 extremely randomized trees.
 - Introduces more randomness than the Random Forest, improving robustness.
 - Serves as one vote in the final ensemble.
- **Voting Classifier (Ensemble)**
 - Role: Combines the three models above into the final detector.
 - Configuration:
 - voting="soft"
 - weights=[1, 1, 1]
 - Function:
 - Combines Random Forest, MLP, and Extra Trees into a single model.
 - Uses soft voting — averaging the predicted probabilities of all three.
 - Produces the final, more reliable prediction.

5.3 Part of Implementation

generate_spoofed_dataset()

This function creates the labeled dataset used by the system. It takes a clean recording of genuine human movement and injects spoofing attacks across multiple attack windows, while keeping the beginning and end of each session normal. Roughly 28% of the records are converted into spoofed samples, reproducing realistic manipulation behaviors such as gradual positional drift, frozen coordinates, replayed paths, and inconsistent satellite or signal readings. It outputs a labeled training dataset, a metadata file, and an attack-window report, ensuring the spoofing patterns are controlled and reproducible.

load_data()

This function represents the first stage of the system: preparing the raw GPS data for machine learning. The dataset is loaded from a CSV file containing the eleven standard bracelet fields (session_id, gps_date, gps_time, latitude, longitude, velocity, course, satellites_in_view, satellites_used, hdop, and label). The label column is mapped so that the value 0 always corresponds to normal GPS data and 1 to spoofed data, guaranteeing consistent labels throughout the analysis. The function also computes the time interval between consecutive readings within each session, preserving the original recording order instead of relying on automatic timestamp sorting.

create_features()

This function performs the movement-consistency feature engineering that distinguishes Part Two. Rather than relying on raw coordinates, it computes the real distance traveled between consecutive points using the Haversine formula and compares it with the reported velocity to produce a speed-residual feature. It also derives the ratio and discrepancy between satellites in view and satellites used, the difference between the reported course and the bearing implied by the movement, acceleration, and changes in the Horizontal Dilution of Precision (HDOP). Finally, rolling-window statistics (mean and standard deviation over a five-record window) are computed for the most informative features, producing a total of thirty-one features.

analyze_data_quality()

The purpose of this function is not to train a model, but to understand the nature of the dataset itself. In cybersecurity applications, extremely high accuracy can be misleading. This function compares key feature values between normal and spoofed records to check whether the classes are clearly separated; if the distributions barely overlap, even simple rules could separate them, so perfect accuracy would not necessarily indicate a robust system. This analysis places the model's performance into context and prevents overestimating its effectiveness in noisier, real-world conditions.

train_and_evaluate()

This function implements the core learning process. It separates the input features from the class labels and applies preprocessing — median imputation, standard scaling, and a Yeo-Johnson power transformation — so all attributes contribute fairly during training. The data is split using a stratified 75/25 split to preserve class balance. Three classifiers are then trained independently — a Random Forest, a Multi-Layer Perceptron (MLP) neural network, and an Extra Trees classifier — and combined into a soft-voting ensemble that averages their probability outputs for a more reliable decision. Performance is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix.

evaluate_time_based_split()

This function provides a more realistic evaluation by training on the earliest records and testing on the latest ones, rather than splitting randomly. This simulates how the system would behave on a continuous GPS stream and helps confirm that performance is not the result of randomly mixed training and test data.

predict_all_and_save()

This function applies the trained ensemble to all available GPS records, using the same preprocessing steps as training to ensure consistency. Numeric predictions are converted into readable labels (normal or spoofed) with confidence scores derived from the

predicted probabilities. The function then saves the predictions, the trained ensemble model, and its preprocessing objects as PKL files, so the model can be reused without retraining.

run_blind_prediction()

This function simulates real deployment. It loads the previously saved model and applies it to a completely separate, unlabeled dataset (with the label column removed), producing predictions without any access to the true answers.

score_blind_predictions()

This function provides the honest, final assessment. It compares the blind predictions against a hidden answer-key dataset containing the true labels and reports the resulting accuracy, confusion matrix, and classification report — confirming how well the system generalizes to genuinely unseen data.

detect_attack_types()

Beyond classifying a record as normal or spoofed, this function explains the likely type of each detected attack. Using rule-based checks on the engineered features, it assigns one of six categories — gradual drag, freeze, replay, fake walking, geofence evasion, and signal manipulation — based on indicators such as a large speed residual, near-zero velocity over an extended period, or an inconsistent course-to-bearing relationship. This adds transparency and helps analysts understand the nature of an attempted manipulation.

5.4 Graphical User Interfaces

Two desktop applications were developed using the CustomTkinter library so the system can be used without the command line. The workflow application separates the process into three clear stages — training a model on a labeled dataset, predicting on an unlabeled dataset, and evaluating predictions against true labels. The dashboard application

performs a one-click analysis and presents visualizations such as feature importance, confusion matrices, and attack-type reports. Both applications reuse the same detection engine, ensuring identical results regardless of the interface used.

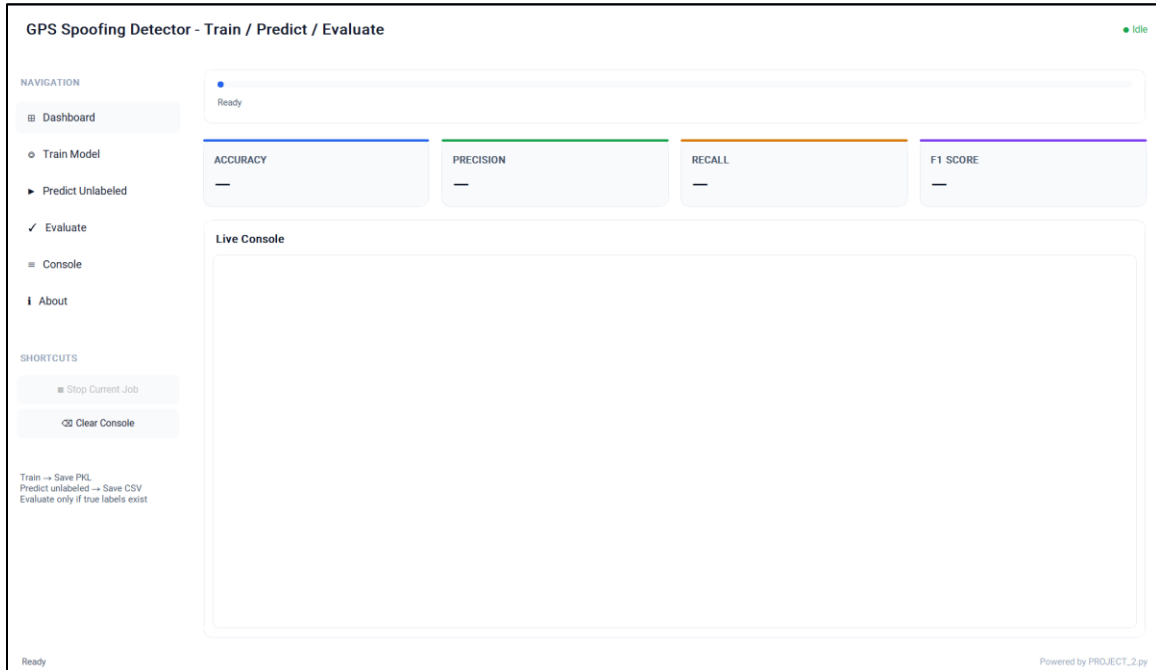


Figure 9 .SpoofGuard Workflow Application Interface

Table 5 .The Two Graphical User Interfaces of the System

Aspect	SpoofGuard Workflow	SpoofGuard Dashboard
Purpose	A controlled, step-by-step interface that follows the full machine-learning workflow.	A one-click interface for fast analysis and rich visual reporting.
Main sections	Dashboard, Train Model, Predict Unlabeled, Evaluate, Console, About.	Dashboard, Console, Results, Attack Report, Blind Test, About.

How it works	The user trains a model on a labeled dataset, predicts on a separate unlabeled dataset, then evaluates the predictions against true labels — each as a distinct stage.	The user loads one labeled dataset and clicks "Run Analysis," which trains, evaluates, and analyzes the data in a single pass.
Key capabilities	Automatic model saving/loading (PKL), input validation, prediction export to CSV, accuracy/precision/recall/F1 and confusion matrix.	Feature-importance chart, model-performance table, three-category attack-type report, confusion matrix, and a separate blind-test view.
Best suited for	Rigorous, reproducible evaluation that mirrors the train/predict/evaluate methodology.	Quick demonstrations and visual interpretation of results and attack types.

Chapter 6: Testing Plan

6.1 Testing

The system was tested using two independent recording sessions from the generated human-movement dataset. A large session of 40,546 records was used for training (29,408 normal and 11,138 spoofed), and a smaller, fully independent session of 7,423 records (5,345 normal and 2,078 spoofed) was reserved exclusively for blind testing. Three complementary evaluation strategies were applied: a stratified 75/25 split, a time-based split, and a blind test on completely unseen data scored against a hidden answer key.

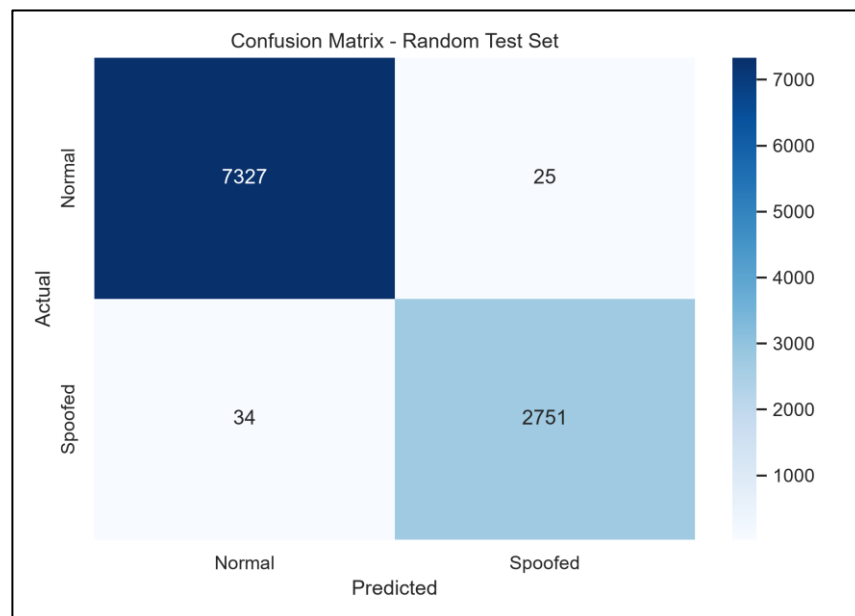


Figure 10 .Confusion Matrix – Stratified Test Set

This confusion matrix shows the model's performance on the 25% held-out test set. The ensemble achieved an accuracy of 99.42%, with precision, recall, and F1-score all close to 99%, demonstrating strong learning on data it did not train on.

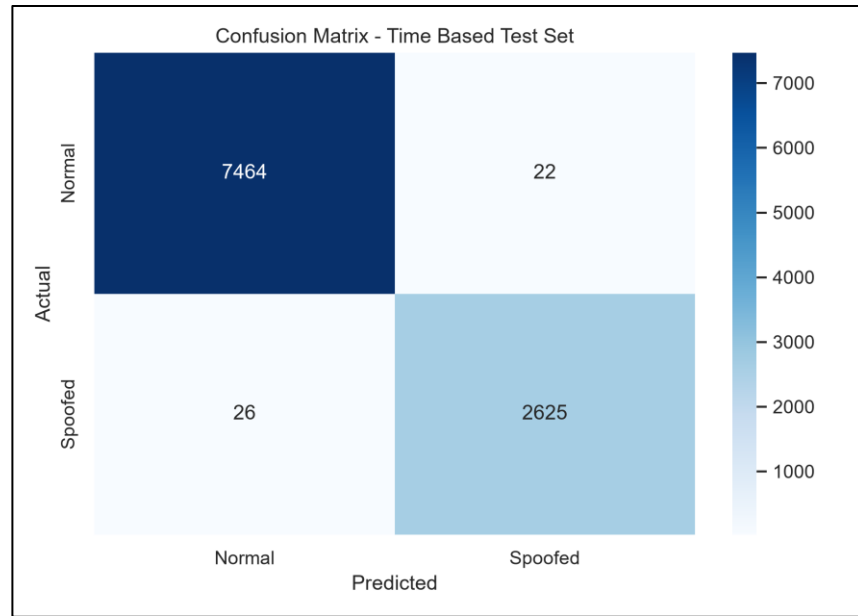


Figure 11 .Confusion Matrix – Time-Based Test Set

This matrix shows performance when the model is trained on the earliest records and tested on the latest ones, simulating a continuous GPS stream. The accuracy of 99.53% confirms the model remains reliable in a realistic, time-ordered setting.

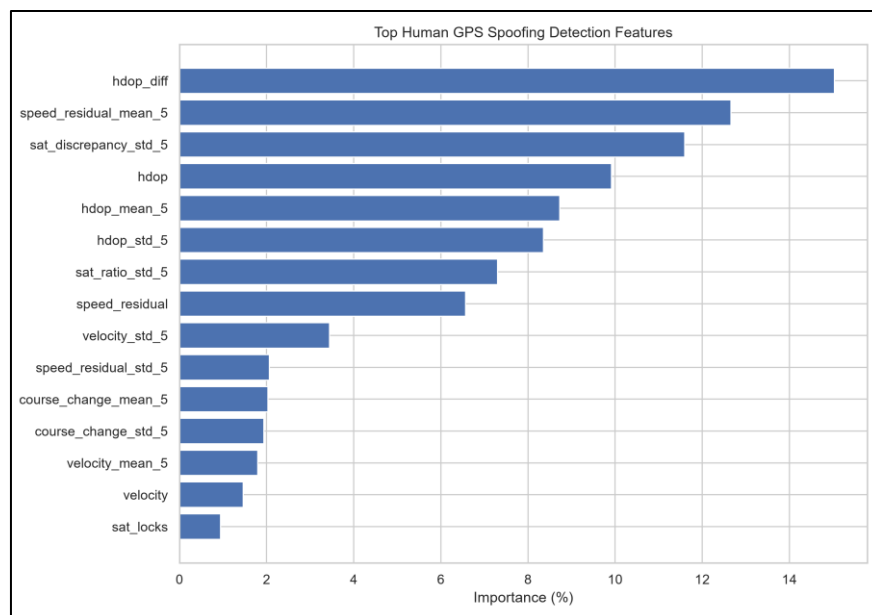


Figure 12 .Feature Importance Graph

This graph presents the relative importance of the engineered features in the Random Forest model. The most influential features are the movement-consistency measures —

particularly the speed residual (the mismatch between reported velocity and the distance actually traveled) together with the satellite-based features — confirming that physical inconsistency is the strongest indicator of spoofing.

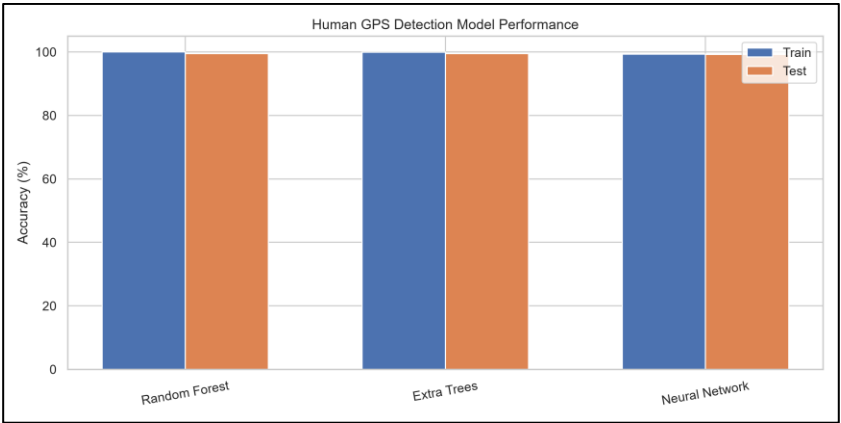


Figure 13 .Model Comparison Graph

This figure compares the training and testing accuracy of the individual models (Random Forest, Multi-Layer Perceptron, and Extra Trees) against the soft-voting ensemble. The ensemble achieves the highest and most consistent performance, demonstrating improved generalization over any single model.

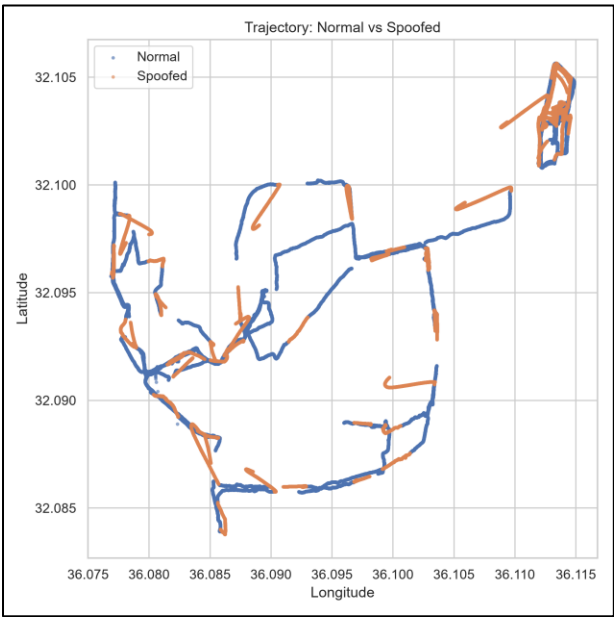


Figure 14 .Normal vs Spoofed

This figure plots the recorded movement path, highlighting normal and spoofed points, and visually illustrates how spoofing distorts the genuine human-movement trajectory.

6.2 Scenarios

To illustrate how the system behaves, two records were taken from the dataset — one labeled 0 (normal) and one labeled 1 (spoofed):

- 0 means the record is normal.
- 1 means the record is fake (spoofed).

The system uses a labeled GPS dataset where these labels serve only as ground truth for training and evaluation. During prediction, the model does not rely on the original label of a record. Instead, it analyzes the engineered movement-consistency features — such as the speed residual (the mismatch between reported velocity and the distance actually traveled), the satellite ratio and discrepancy, the course-to-bearing difference, and changes in HDOP — to decide whether the record is normal or spoofed.

As a result, a record originally labeled 0 could still be classified as fake if its feature values resemble spoofing behavior. This is theoretically possible in any machine-learning system; however, in this dataset such misclassification is rare because the engineered features separate normal and spoofed movement clearly, which leads to very high accuracy.

Table 6 . Records Before Testing

Field	Record 1	Record 2
velocity	0.097	1.26
course	0.0	346.87
satellites_in_view	10	11
satellites_used	5	8
hdop	1.81	1.12
label (ground truth)	0 (normal)	1 (spoofed)

The first record was predicted Normal, which is correct, and the second record was predicted Spoofed, which is also correct. For the spoofed record, the system additionally applied its attack-type module and explained the manipulation as one of the six recognized categories (gradual drag, freeze, replay, fake walking, geofence evasion, or signal manipulation), based on the characteristic indicators present in the record.

Chapter 7: Conclusion and Results

7.1 Summary of Accomplished Project

Objectives: The project aimed to enhance the security of Electronic Monitoring (EM) bracelets by studying GPS spoofing attacks and developing a detection approach capable of identifying manipulated GPS behavior. The main objective was to detect inconsistencies between GPS features such as location, velocity, course, satellite count, satellite locks, and HDOP, which may indicate spoofed or abnormal GPS records.

Methodology: The project first involved collecting real human-movement GPS data using a Raspberry Pi and Neo-6M GPS module. The collected data was stored in CSV format and organized using session IDs to separate different collection periods. After that, realistic GPS spoofing attacks were simulated on part of the normal data, including freeze, replay, and geofence evasion attacks. The generated labeled dataset was then used to train and test a machine learning-based detection model. Movement-based and signal-quality features were extracted, and multiple models were combined using an ensemble approach to improve detection performance.

Outcomes: The project successfully produced a realistic human-movement GPS dataset and a labeled spoofed dataset suitable for training and evaluation. A GPS spoofing detection system was developed to classify GPS records as normal or spoofed based on extracted features. In addition, the detection code was integrated into a graphical user interface (GUI), allowing users to train the model, predict unlabeled GPS data, evaluate predictions using true labels, and display performance results such as accuracy, precision, recall, F1-score, and confusion matrix. The final system provides a practical approach for detecting GPS spoofing attempts and improving the reliability of GPS-based electronic monitoring systems.

7.2 Future Work

Future work can focus on improving the system by collecting a larger and more diverse GPS dataset from different environments, movement patterns, and weather conditions.

This would help improve the model's generalization and make the detection system more reliable in real-world scenarios. The system can also be extended to support live GPS data streaming directly from the Raspberry Pi and Neo-6M module, allowing near real-time spoofing detection instead of relying only on offline CSV files.

Another important future improvement is testing the system with real electronic monitoring bracelet data, if such data becomes available from official authorities. This would allow better validation of the model in an actual EM environment. In addition, more spoofing scenarios can be added, and the GUI can be enhanced with real-time alerts, better visualization, and automatic report generation. Finally, the system could be integrated with a monitoring center or server-side platform to support practical deployment in electronic monitoring systems.

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